

Name

M#

Models

Review $y' = A y + b$

Chapter 4:

Driven Mass-Spring system

$$m y'' + c y' + k y = f_0 \sin(\omega t)$$

The ODE

$$m y'' + c y' + k y = f_0 \sin(\omega t)$$

is the standard mass-spring model. The mass and spring are defined by the positive constants mass m , linear damping coefficient c , and spring constant k . The sinusoidal external force is described by the magnitude f_0 and the angular frequency ω . The damping c is usually small and we are interested in the response of the system to a broad range of angular frequencies from slow $\omega \ll 1$ to fast $\omega \gg 1$. We are going to see that there is a single angular frequency ω_0 near which small forces can produce outsized motions.

Complementary Solution

The roots

$$\lambda_{\pm} = \frac{-c \pm \sqrt{c^2 - 4km}}{2m}$$

of the characteristic polynomials

$$m \lambda^2 + c \lambda + k = 0$$

give the complementary solution of

$$m y'' + c y' + k y = 0.$$

If $c = 0$ the roots are pure imaginary

$$\lambda_{\pm} = \frac{-0 \pm \sqrt{0^2 - 4km}}{2m} = \pm i \sqrt{\frac{k}{m}}.$$

writing

$$\omega_0 = \sqrt{\frac{k}{m}}$$

for the undamped (because $c = 0$) natural frequency we see that both terms in the complementary

solution

$$c_1 \cos(\omega_0 t) + c_2 \sin(\omega_0 t)$$

oscillate at the natural frequency without decaying.

If $\frac{c}{2m} < \omega_0$ the roots are complex

$$\lambda_{\pm} = \frac{-c \pm \sqrt{c^2 - 4km}}{2m} = -\frac{c}{2m} \pm \frac{\sqrt{c^2 - 4km}}{\sqrt{(2m)^2}} = -\frac{c}{2m} \pm i \sqrt{\frac{k}{m} - \left(\frac{c}{2m}\right)^2}$$

writing

$$\omega_c = \sqrt{\frac{k}{m} - \left(\frac{c}{2m}\right)^2} \leq \omega_0$$

makes it easy to see that both terms

$$e^{-\left(\frac{c}{2m}\right)t} (c_1 \cos(\omega_c t) + c_2 \sin(\omega_c t))$$

in the complementary solution decay exponentially like

$$e^{-\left(\frac{c}{2m}\right)t}$$

while oscillating slower than the undamped ω_0 .

If $\frac{c}{2m} > \omega_0$ the roots

$$\lambda_+ = \frac{-c + \sqrt{c^2 - 4km}}{2m} \text{ and } \lambda_- = \frac{-c - \sqrt{c^2 - 4km}}{2m}$$

are real with $\lambda_- < \lambda_+ < 0$ and both terms in the complementary solution

$$c_1 e^{\lambda_+ t} + c_2 e^{\lambda_- t}$$

decay exponentially as $t \rightarrow \infty$ with the slower decay rate given by

$$\lambda_+ = \frac{-c + \sqrt{c^2 - 4km}}{2m}$$

so

$$c_1 e^{\lambda_+ t} + c_2 e^{\lambda_- t} \approx c_1 e^{\lambda_+ t} \text{ as } t \rightarrow \infty$$

These three cases are called:

- Undamped ($c = 0$)

Oscillations with frequency $\omega_0 = \sqrt{\frac{k}{m}}$ that do not decay.

- Underdamped ($\frac{c}{2m} < \omega_0$)

Oscillations with frequency $\omega_c = \sqrt{\frac{k}{m} - \left(\frac{c}{2m}\right)^2}$ that decay exponentially like $e^{-\left(\frac{c}{2m}\right)t}$

- Overdamped ($\frac{c}{2m} > \omega_0$)

No oscillations just exponential decay

Particular Solution

The undetermined coefficient form for

$$m y'' + c y' + k y = f_0 \sin(\omega t)$$

is

$$y_p = a_1 \cos(\omega t) + a_2 \sin(\omega t)$$

with derivatives

$$\dot{y}_p = -\omega a_1 \sin(\omega t) + \omega a_2 \cos(\omega t)$$

$$y_p'' = -\omega^2 a_1 \cos(\omega t) - \omega^2 a_2 \sin(\omega t)$$

Substituting into the ODE gives

$$m(-\omega^2 a_1 \cos(\omega t) - \omega^2 a_2 \sin(\omega t)) + \\ c(-\omega a_1 \sin(\omega t) + \omega a_2 \cos(\omega t)) + \\ k(a_1 \cos(\omega t) + a_2 \sin(\omega t)) = f_0 \sin(\omega t)$$

Matching the coefficients of $\sin(\omega t)$ gives

$$m(-\omega^2 a_2 \sin(\omega t)) + \\ c(-\omega a_1 \sin(\omega t)) + \\ k(a_2 \sin(\omega t)) = f_0 \sin(\omega t)$$

which implies

$$(\text{Sin-Eq}) \quad -m \omega^2 a_2 - c \omega a_1 + k a_2 = f_0.$$

Matching the coefficients of $\cos(\omega t)$ gives

$$m(-\omega^2 a_1 \cos(\omega t)) + \\ c(\omega a_2 \cos(\omega t)) + \\ k(a_1 \cos(\omega t)) = 0$$

which implies

$$(\text{Cos-Eq}) \quad -m \omega^2 a_1 + c \omega a_2 + k a_1 = 0.$$

Combining the equations and organizing gives

$$(\text{Sin-Eq}) \quad (k - m \omega^2) a_2 - c \omega a_1 = f_0$$

$$(\text{Cos-Eq}) \quad (k - m \omega^2) a_1 + c \omega a_2 = 0$$

Reordering the terms in the sine equation gives

$$-c \omega a_1 + (k - m \omega^2) a_2 = f_0 \\ (k - m \omega^2) a_1 + c \omega a_2 = 0$$

which is simply

$$\begin{pmatrix} -c \omega & k - m \omega^2 \\ k - m \omega^2 & c \omega \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} f_0 \\ 0 \end{pmatrix}$$

As always, call the matrix A then flashback to linear algebra. Remember the solution to

$$A \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} f_0 \\ 0 \end{pmatrix}$$

can be large if the determinant of A is small! We can actually work out (by hand if we had to) how large the solution is

$$\left\| \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \right\|^2 = \frac{f_0^2}{k^2 + c^2 \omega^2 - 2 k m \omega^2 + m^2 \omega^4}$$

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In[74]:= Clear[m, k, c, ω]
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$$A = \begin{pmatrix} -c \omega & k - m \omega^2 \\ k - m \omega^2 & c \omega \end{pmatrix};$$

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a = LinearSolve[A, {1, 0}];
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Simplify[a.a]
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Out[77]=
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$$\frac{1}{k^2 + c^2 \omega^2 - 2 k m \omega^2 + m^2 \omega^4}$$

we can tidy up the mess on the bottom quite a bit. Divide top and bottom by m^2 to get

$$\frac{1}{k^2 + c^2 \omega^2 - 2 k m \omega^2 + m^2 \omega^4} = \frac{1/m^2}{(k/m)^2 + (c/m)^2 \omega^2 - 2 (k/m) \omega^2 + \omega^4}$$

then recognize $\omega_0^2 = k/m$ to get

$$\frac{1}{(k/m)^2 + (c/m)^2 \omega^2 - 2 (k/m) \omega^2 + \omega^4} = \frac{1}{(\omega_0^2)^2 + (c/m)^2 \omega^2 - 2 \omega_0^2 \omega^2 + \omega^4}$$

then collect terms and complete the square to get

$$\frac{1}{\omega_0^4 - 2 \omega_0^2 \omega^2 + \omega^4 + (c/m)^2 \omega^2} = \frac{1}{(\omega^2 - \omega_0^2)^2 + (c/m)^2 \omega^2}$$

Putting this all together we get

$$\left\| \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \right\|^2 = \frac{f_0^2/m^2}{(\omega^2 - \omega_0^2)^2 + (c/m)^2 \omega^2}$$

General Solution and Interpretation

$$y'' + \frac{c}{m} y' + \frac{k}{m} y = \frac{f_0}{m} \sin(\omega t)$$

The general solution for the physically interesting underdamped case $\frac{c}{2m} < \omega_0 = \sqrt{\frac{k}{m}}$ is

$$y = e^{-\frac{c}{2m} t} (c_1 \cos(\omega_c t) + c_2 \sin(\omega_c t)) + a_1 \cos(\omega t) + a_2 \sin(\omega t)$$

where $\left\| \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \right\|^2 = \frac{f_0^2/m^2}{(\omega^2 - \omega_0^2)^2 + (c/m)^2 \omega^2}$. In other words, the response is an exponentially decaying transient with a steady state response that can be large if the damping is small and the driving frequency is near the natural frequency of the system.

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In[78]:= m = 1;
ω0 = 1;
Manipulate[
  LogPlot[ $\left\{\frac{1}{(\omega^2 - \omega_0)^4 + \left(\frac{c}{m}\right)^2 \omega^2}, \frac{1}{\left(\frac{c}{m}\right)^2 \omega_0^2}\right\}$ , {ω, 0, 2 ω0},
    PlotRange → All, PlotPoints → 100, GridLines → Automatic],
  {{c, 0.001}, 0, 0.1}]

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Out[80]=

